FLATNESS OF A MODULE AND OF ITS BASE CHANGE ALONG A LOCAL DIAGRAM, BOTH DIRECTIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Preservation of flatness by base change. Invoked in the counted proof at lines 24426-24428. Gives the first direction directly and is reused after localising.
- Flatness is preserved by localisation. Used in the counted proof at lines 24429-24431. Identifies the base-changed module with a localisation of the tensor product, which is the form in which flatness is known.

Source reference. Original lemma 00HG: sources/stacks-algebra/pinned-original-source.tex, source lines 8936–8936. Navigation pointer for \ref{item-f-ideal} used inside the counted span (line 24441); the argument that consumes it is retained in the span.

Source reference. Original lemma 00HD: sources/stacks-algebra/pinned-original-source.tex, source lines 8925–8925. Navigation pointer for \ref{lemma-flat} used inside the counted span (line 24441); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00HI: sources/stacks-algebra/pinned-original-source.tex, source lines 9106–9106. Navigation pointer for \ref{lemma-flat-base-change} used inside the counted span (line 24428); the argument that consumes it is retained in the span.

Source reference. Original lemma 00HT: sources/stacks-algebra/pinned-original-source.tex, source lines 9413–9413. Navigation pointer for \ref{lemma-flat-localization} used inside the counted span (line 24432); the argument that consumes it is retained in the span.

Source reference. Original lemma 00HR: sources/stacks-algebra/pinned-original-source.tex, source lines 9401–9401. Navigation pointer for \ref{lemma-local-flat-ff} used inside the counted span (line 24439); the argument that consumes it is retained in the span.

Lemma 0.1. *Let*

$$\begin{array}{ccc} S & \longrightarrow & S' \\ \uparrow & & \uparrow \\ R & \longrightarrow & R' \end{array}$$

be a commutative diagram of local homomorphisms of local rings. Assume that S' is a localization of the tensor product $S \otimes_R R'$. Let M be an S-module and set $M' = S' \otimes_S M$.

(1) *If M is flat over R then M' is flat over R'.*
(2) *If M' is flat over R' and $R \to R'$ is flat then M is flat over R.*

In particular we have

(3) *If S is flat over R then S' is flat over R'.*
(4) *If $R' \to S'$ and $R \to R'$ are flat then S is flat over R.*

Proof. Proof of (1). If M is flat over R, then $M \otimes_R R'$ is flat over R' by Lemma 00HI. If $W \subset S \otimes_R R'$ is the multiplicative subset such that $W^{-1}(S \otimes_R R') = S'$ then $M' = W^{-1}(M \otimes_R R')$. Hence M' is flat over R' as the localization of a flat module, see Lemma 00HT part (5). This proves (1) and in particular, we see that (3) holds.

Proof of (2). Suppose that M' is flat over R' and $R \to R'$ is flat. By (3) applied to the diagram reflected in the northwest diagonal we see that $S \to S'$ is flat. Thus $S \to S'$ is faithfully flat by Lemma 00HR. We are going to use the criterion of Lemma 00HD (00HG) to show that M is flat. Let $I \subset R$ be an ideal. If $I \otimes_R M \to M$ has a kernel, so does $(I \otimes_R M) \otimes_S S' \to M \otimes_S S' = M'$. Note that $I \otimes_R R' = IR'$ as $R \to R'$ is flat, and that

$$(I \otimes_R M) \otimes_S S' = (I \otimes_R R') \otimes_{R'} (M \otimes_S S') = IR' \otimes_{R'} M'.$$

From flatness of M' over R' we conclude that this maps injectively into M'. This concludes the proof of (2), and hence (4) is true as well. □